



HashSet in C# Tutorial

Tutorial on using HashSet collections in C# to speed up searches in applications.
<http://dgit.in/nov20-67>



NVIDIA Jetson Nano 2GB

Powered by CUDA-X, the single-board developer kit is aimed towards robotics enthusiasts. <http://dgit.in/nov20-68>

lowering the barrier of access. Putting together a working computer with Raspberry Pi needed just a MicroSD card onto which one could install Raspbian, the official Raspberry Pi operating system. A fork of Debian, Raspbian was very lean and would provide users with a handy GUI which made it much easier to use. And since the Raspberry Pi had a much powerful Broadcom SoC, IoT projects that needed a lot more compute power could easily be prototyped on a Pi.

Early adopters even used the Raspberry Pi as media servers since you could easily play Full HD content



off of the Pi and even install a media management software such as Plex on it. Before Android TVs came onto the scene, you had Raspberry Pi media boxes making TVs “smart”. For compact form factors, Raspberry Pi has the Pi Zero board. The only complaint from the maker community regarding Raspberry Pi was that the project was proprietary. Nevertheless, the Pi scene is one of the largest IoT scenes in the world.

PYCOM

Pycom boards are relatively new since they only started out about five years ago. Their boards support multiple network protocols which is attractive to IoT developers since each network protocol has its own strengths and weaknesses and a single line of dev boards that support multiple protocols makes it easy to switch up between these different network protocols with minimal modifications to the code base. Speaking of which, Pycom boards are designed for MicroPython which is a full Python compiler written in C

and mostly compatible with Python 3. Essentially, it's Python that's designed to run on microcontrollers and gives developers access to low-level hardware. Given how Python is



the hottest language on the block, it makes it very easy for developers to get started with Pycom boards.

The boards are pin-compatible and based on the ESP32 microcontrollers and folks can get WiPy, SiPy, LoPy, GPy and FiPy boards based on whichever network protocol they wish to implement in their IoT projects. The same has been enumerated below:

- WiPy - Wi-Fi, Bluetooth
- SiPy - Sigfox, Wi-Fi, Bluetooth
- LoPy - LoRa, Wi-Fi, Bluetooth
- GPy - LTE CAT M1/NB1, Wi-Fi, Bluetooth
- FiPy - Dual LTE-M, Sigfox, LoRa, Wi-Fi, Bluetooth

It goes without saying that FiPy is the most popular of the lot for prototyping given that it supports all the network protocols. There are also expansion boards available to extend the features and enable even more interfaces such as USB and PoE (Power over Ethernet).

ADAFRUIT FEATHER MO

Powered by an ATSAM21G18 ARM Cortex MO SoC, the same that powers Arduino Zero, the Adafruit Feather MO is a lightweight dev board for compact IoT projects. The SoC has 256K of flash memory which is much more than the Atmega328 which is used by the Arduino Uno R3. Also, it has 32K of RAM which is 16x as much as the Atmega328. The board uses a MicroUSB port for interfacing so it's convenient to get started since you don't need to attach an intermediary FTDI chip. They've added a small connector for plugging in a 3.7V LiPo battery making IoT projects built using

the Feather MO to be portable. Also, they've added a 100mA LiPo charger with a status LED onto the board.

The main SoC breakout board is very small so the rest of the PCB consists of through-holes which can be used as a prototyping area. The pins are arranged along the periphery and are clearly labelled so you don't have to worry about figuring out which is the prototyping area and which are the pinouts for the SoC. All you need to get started are two berg strips that need to be soldered along both the longer edges of the board.

The Feather MO uses the Arduino IDE but is not supported out of the box. You will have to import the Adafruit support packages by adding the URI for Adafruit Arduino boards. Also, not every Arduino sketch



written for the Arduino Zero will work right away. You might have to find-and-replace some elements of the code and then you should be set.

PARTICLE IOT

Particle IoT is also a relatively new company which offers full-stack IoT kits, which is a way of saying that they provide you with all the hardware and software required to get your project started. This also means that the folks over at Particle IoT only have to focus on a restrictive set of hardware rather than opening it up to everything under the sun. A great aspect of Particle is that they provide access to Particle's cloud services so you can build a cloud-enabled IoT project using their API. They've already partnered with several 2G/3G service providers in different regions to incorporate Particle cloud into their services. At last count, they'd partnered with service providers in